DameLibraries: Using Python Libraries from Tests

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1 Introduction

Python is a winner in the trend market of programming languages in this moment. Although the sintaxis is so comfortable to the programmers, the clue is in the modern libraries for system administrators, scientifics and with low importance for web programmers.

You can see the Tiobe Index:

Jan 2021	Jan 2020	Change	Programming Language	Ratings	Change
1	2	⌃	C	17.38%	+1.61%
2	1	⌄	Java	11.96%	-4.93%
3	3		Python	11.72%	+2.01%
4	4		C++	7.56%	+1.99%
5	5		C#	3.95%	-1.40%
6	6		Visual Basic	3.84%	-1.44%
7	7		JavaScript	2.20%	-0.25%
8	8		PHP	1.99%	-0.41%
9	18	⌃⌃	R	1.90%	+1.10%
10	23	⌃⌃	Groovy	1.84%	+1.23%
11	15	⌃⌃	Assembly language	1.64%	+0.76%
12	10	⌄	SQL	1.61%	+0.10%
13	9	⌄⌄	Swift	1.43%	-0.36%
14	14		Go	1.41%	+0.51%

You can check that Python is growing fast and reaching the most important positions in the market. The third position in January of 2021.

You can use or don't use a test system such as nose or pytest, but all people need test the code in the programming. After to write software you must introduce inputs in your code and verify if the output is ok.

The advantage using a test system is generate automation in the process about inputs, code being tested and outputs. Each test is evaluated with asserts, that's comparations between the result that you deserve and the result obtained. For instance,

```
def test_models_lda(self):
    X = np.array([[-1, -1], [-2, -1], [-3, -2],
                  [1, 1], [2, 1], [3, 2]])
    y = np.array([1, 1, 1, 2, 2, 2])
    clf = LinearDiscriminantAnalysis()
    clf.fit(X, y)
    self.assertEqual(clf.predict([[-1, -1]]), 1)
```

So, assertEqual is the comparision and the rest is the code to be verified.

To become a professional in Python you must learn a good bunch of libraries. So you can practice the libraries reading tutorials, python notebooks, or you could write tests about this libraries and perhaps to share these tests in some software repository.

If you save all test code about your python libraries in one or various software repositories you can test all code with a command. The python libraries can suffer chances, but you have the control about the use that you do about your libraries.

The goal of this book is to teach Python Libraries from the tests.

2 Installation

You can create a python virtual environment and to install damelibraries with python package:

```
$ python3 -m venv /tmp/venv
$ cd /tmp/venv
$ source bin/activate
$ pip install --upgrade pip
$ pip3 install damelibraries
$ cd lib/python3.5/site-packages/damelibraries
```

Now you can execute tests to check different libraries:

```
$ cd /tmp/venv/lib/python3.8/site-packages/damealgorithms
$ nosetests3 tests

$ cd /tmp/venv/lib/python3.8/site-packages/dameformats
$ nosetests3 tests

$ cd /tmp/venv/lib/python3.8/site-packages/damenltk
$ nosetests3 tests

$ cd /tmp/venv/lib/python3.8/site-packages/damenumpy
$ nosetests3 tests

$ cd /tmp/venv/lib/python3.8/site-packages/damepandas
$ nosetests3 tests

$ cd /tmp/venv/lib/python3.8/site-packages/damescikit
$ nosetests3 tests

$ cd /tmp/venv/lib/python3.8/site-packages/damescipy
$ nosetests3 tests
```

3 DameBasics

Before, to start to learn python libraries you can remember python sintaxis about control structures such as if, while, for, ... and datastructures such as lists, dictionaries, tuples, sets, dates, ...

In DameBasics you can find some source snippets about it and execute it with:

```
$ cd /tmp/venv/lib/python3.8/site-packages/damebasics
$ pytest-3 tests
```

or

```
$ cd /tmp/venv/lib/python3.8/site-packages/damebasics
$ nosetests3 tests
```

4 DameFormats

Json, csv, xml are very important format files. This chapter is about how to use python with these formats presenting some snippets:

```
$ cd /tmp/venv/lib/python3.8/site-packages/dameformats
$ nosetests3 tests/test_damecsv.py
$ nosetests3 tests/test_damejson.py
$ nosetests3 tests/test_damexml.py
```

5 DameScraping

There are several libraries for to do scraping. The most standard and important library is request. I suggest you to use this library if you have doubts about what library to choose.

There are another libraries with modern features such as MechanicalSoup.

Another idea is libraries such as newspaper where you can integrate NLTK in a specific domain (newspaper).

Now you can check the source of the libraries.

```
$ cd /tmp/venv/lib/python3.8/site-packages/damescraping
$ nosetests3 test/test_requests.py
$ nosetests3 test/test_mechanicalsoup.py
$ nosetests3 test/test_newspaper.py
```

6 DameNumpy

Numpy is a library about algebra used as dependency in a lot of python programs. We have classified the tests in basics, datatypes and maths. Basics is related to the most common uses about the library (sum vectors, transpose matrix, generate eye matrix, ...). Datatypes is about specific uses about the numpy datatypes that perhaps you need practice more. And maths is about maths operations. You can execute the test with:

```
$ cd /tmp/venv/lib/python3.8/site-packages/damenumpy
$ nosetests3 tests/test_basics.py
$ nosetests3 tests/test_datatypes.py
$ nosetests3 tests/test_maths.py
```

7 DamePandas

Pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language.

```
$ cd /tmp/venv/lib/python3.8/site-packages/damepandas
$ nosetests3 test/test_create.py
$ nosetests3 test/test_json.py
$ nosetests3 test/test_operators.py
$ nosetests3 test/test_utils.py
```

In test_create.py is being created tests for create datastructures. In test_json.py we practice some json utils in Pandas. In test_operators.py and test_utils.py there are more functions related to Pandas such as read_csv, concat, shape, ...

8 DameScipy

Scipy is a popular library focused on maths for science and computing released with BSD license. You can check the tests now:

```
$ cd /tmp/venv/lib/python3.8/site-packages/damescipy
$ nosetest3 tests
```

In test_basics.py you can learn useful maths such as to integrate, to interpolate, the distance between vectors, the polinomial roots, ...

In null hypothesis significance testing, the p-value is the probability of obtaining test results at least as extreme as the results actually observed, under the assumption that the null hypothesis is correct. In test_pvalues.py you can learn how to use it in python.

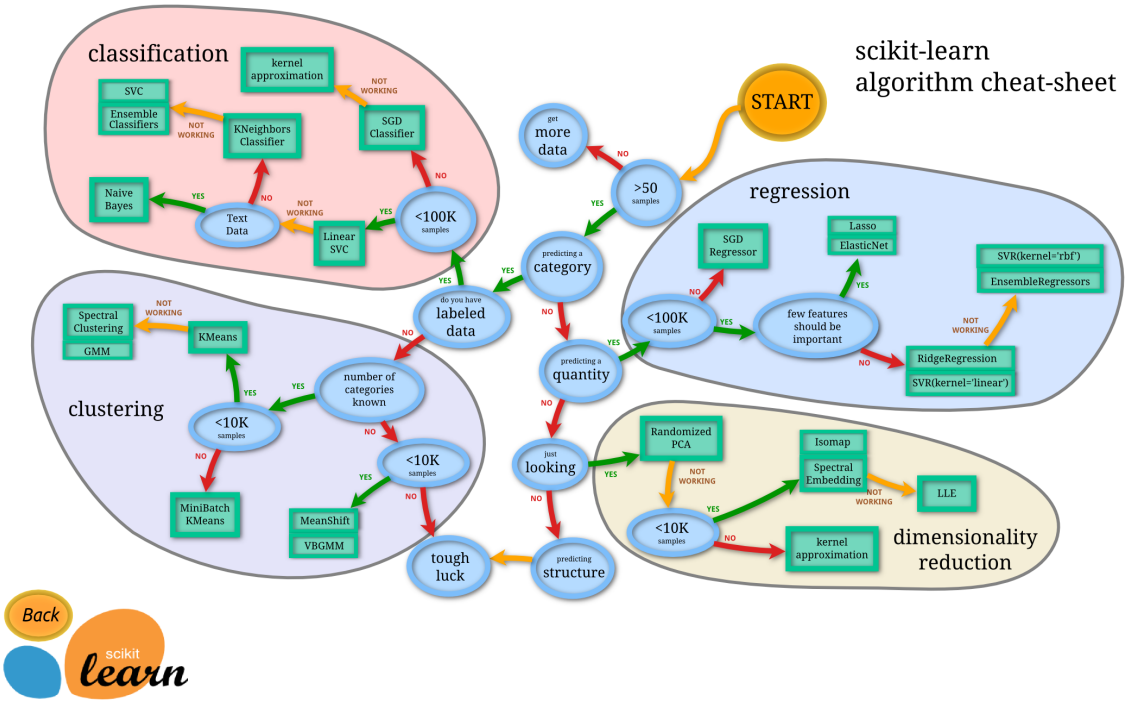
You can learn to use statistical distributions with test_stats.py.

9 DameScikit

Scikit is a popular library about Machine Learning (ML) released with BSD. In this chapter we are going to learn to apply algorithms to the different tasks in ML.

- Classification
- Clustering
- Clustering
- Regression
- Dimensionality Reduction

Take a look to the next diagram:



Classification and Clustering is about to learn to decide elements in groups (to classify). In Classification we have a training set, so we are doing supervised learning. In clustering we have not a training set, so we are doing unsupervised learning.

Dimensionality reduction is about to determine the most important factors or features to make the machine learning.

Regression is about to decide a quantity from a set of training examples.

In DameScikit we have a lot of examples about to apply machine learning models on a simple way:

```
$ nosetests3 tests/test_models.py
```

Take a look to a single test. Each array in the vector X (input) has an element in the vector Y (output). So [-1, -1] has the representation 1 and [2, 1] has the representation 2.

Later, we create an object LinearDiscriminantAnalysis (the model). Next, the model is fitted with inputs and outputs (train the model). Finally, we can predict inputs.

```
X = np.array([[-1, -1], [-2, -1], [-3, -2], [1, 1], [2, 1], [3, 2]])
y = np.array([1, 1, 1, 2, 2, 2])
clf = LinearDiscriminantAnalysis()
clf.fit(X, y)
self.assertEqual(clf.predict([[-1, -1]]), 1)
```

That's the most important, but Scikit is a long library with a lot of interesting features. You can check more tests:

```
$ nosetests3 tests
```

10 DameNLTK

NLTK is an acronym about Natural Language ToolKit.

A summary of tasks about Natural Language Processing could be:

- Sentiment Analysis
- Detect Gender
- Sentence Similarity
- Text Summary
- Classify Documents
- Manage Words
- Gramatical Trees
- Extract Keywords
- Disambiguation

You could be in the path of nltk so:

```
$ cd /tmp/venv/
$ source bin/activate
(venv)$ pip3 install damenltk
$ cd /tmp/venv/lib/python3.8/site-packages/damenltk
$ nosetest3 test
```

10.1 Sentiment Analysis

Sentiment analysis is applied in a popular way in Twitter with python libraries such as tweepy. Finally, it is a classification task about words with a label (the sentiment). That could be implemented with nltk. Take a look to the test:

```
import unittest
from nltk.classify import SklearnClassifier
class TddInPythonExample(unittest.TestCase):

    def test_sentiment_variable_create_returns_correct_result(self):
        pos_tweets = [('I love this car', 'positive'),
                      ('This view is amazing', 'positive'),
                      ('I feel great this morning', 'positive'),
                      ('I am so excited about the concert', 'positive'),
                      ('He is my best friend', 'positive'),
                      ('This movie was great', 'positive'),
                      ('This movie was not pathetic', 'positive')]

        neg_tweets = [('I do not like this car', 'negative'),
                      ('This view is horrible', 'negative'),
                      ('I feel tired this morning', 'negative'),
                      ('I am not looking forward to the concert',
                      'negative'),
                      ('He is my enemy', 'negative'),
```

```
                    ('This is a pathetic movie', 'negative')]

        tweets_with_sentiment = []
        for (tweet, sentiment) in pos_tweets + neg_tweets:
            filtered_tweet_words = [word.lower()
                                    for word in tweet.split()
                                    if len(word) >= 3]
            tweets_with_sentiment.append((filtered_tweet_words,
                                          sentiment))

        self.assertEqual([(['love', 'this', 'car'], 'positive'),
                          (['this', 'view', 'amazing'], 'positive'),
                          (['feel', 'great', 'this', 'morning'],
                          'positive'),
                          (['excited', 'about', 'the', 'concert'],
                          'positive'),
                          (['best', 'friend'], 'positive'),
                          (['this', 'movie', 'was', 'great'],
                          'positive'),
                          (['this', 'movie', 'was', 'not', 'pathetic'],
                          'positive'),
                          (['not', 'like', 'this', 'car'], 'negative'),
                          (['this', 'view', 'horrible'], 'negative'),
                          (['feel', 'tired', 'this', 'morning'],
                          'negative'),
                          (['not', 'looking', 'forward', 'the', 'concert'],
                          'negative'),
                          (['enemy'], 'negative'),
                          (['this', 'pathetic', 'movie'], 'negative')],
                          tweets_with_sentiment)
```

You can execute this test with:

```
$ nosetest3 test/test_sentiment.py
```

10.2 Detect Gender

The gender detection from the name is a NLP task. So the input is one or more word (name and perhaps surname) and the output is a gender male, female, ...

NLTK brings a good list of names in the nltk.corpus. So you can classify names with two ways:

- Querying single files
- Supervised classifier

The python method using single files would be:

```
    def gender_name(self, name):
        gender = 'undefined'
        if name in names.words('male.txt'):
```

```
            gender = 'male'
        if name in names.words('female.txt'):
            if (gender == 'male'):
                gender = 'both'
            else:
                gender = 'female'
        return gender
```

And the python test would be:

```
    def test_gender_name_method_returns_correct_result(self):
        dn = DameNLTK()
        self.assertTrue(
                dn.gender_name("Neo"),
                "male")
        self.assertTrue(
                dn.gender_name("Trinity"),
                "female")
        self.assertTrue(
                dn.gender_name("Andrea"),
                "both")
```

The python methods using machine learnig would be:

```
    def gender_features(self, word):
        return 'last_letter': word[-1]

    def gender_classifier(self):
        labeled_names = (
             [(name, 'male') for name in names.words('male.txt')] +
             [(name, 'female') for name in names.words('female.txt')])

        featuresets = [(self.gender_features(n), gender)
                    for (n, gender) in labeled_names]
        train_set, test_set = featuresets[500:], featuresets[:500]
        classifier = nltk.NaiveBayesClassifier.train(train_set)
        return classifier
```

And the python tests would be:

```
    def test_gender_classifier_method_returns_correct_result(self):
        dn = DameNLTK()
        classifier = dn.gender_classifier()
        self.assertTrue(classifier.classify(dn.gender_features("Neo")),
                        "male")
        self.assertTrue(classifier.classify(dn.gender_features("Trinity")),
                        "female")
```

You can use an advanced software to detect gender such as Damegender with many features:

```
$ pip3 install damegender[all]
```

You can learn to use the software with tests oriented to the user (bash files) or tests oriented to the developer (python tests).

```
$ python3 -m venv /tmp/damegender/
$ cd /tmp/damegender/
$ source bin/activate
$ pip3 install damegender[all]
$ ./testsbycommands.sh
maindavid test is ok
mainjesus test is ok
mainines test is ok
mainalex test is ok
mainandrea test is ok
mainjesusmaria test is ok
mainjosemaria test is ok
mainelenaluciahelena test is ok
mainjuliaus test is ok
mainjuliauk test is ok
mainjuliauy test is ok
[...]

$ nosetest3 test
[...]
```

Execute these tests could be long time, but you can execute a single file test, for example, you can execute test/test_dame_utils.py:

```
$ nosetest3 test/test_dame_utils.py
```

Or you can execute a single test in a file with:

```
$ nosetests3 test/test_dame_utils.py:TddInPythonExample.test_string2array
```

10.3 Sentence Similarity

Remembering to W H Gomaa and A A Fahmy in "A Survey of Text Similarity Approaches":

Finding similarity between words is a fundamental part of text similarity which is then used as a primary stage for sentence, paragraph and document similarities. Words can be similar in two ways lexically and semantically. Words are similar lexically if they have a similar character sequence. Words are similar semantically if they have the same thing, are opposite of each other, used in the same way, used in the same context and one is a type of another.

In the python method sentence_similarity we apply similarity lexical. So, it's dividing the text in two vectors (vector1 and vector2) and we apply the cosine distance:

```
    def sentence_similarity(self, sent1, sent2, stopwords=None):
        if stopwords is None:
            stopwords = []

        sent1 = [w.lower() for w in sent1]
        sent2 = [w.lower() for w in sent2]
```

```
        all_words = list(set(sent1 + sent2))

        vector1 = [0] * len(all_words)
        vector2 = [0] * len(all_words)

        # build the vector for the first sentence
        for w in sent1:
            if w in stopwords:
                continue
            vector1[all_words.index(w)] += 1

        # build the vector for the second sentence
        for w in sent2:
            if w in stopwords:
                continue
            vector2[all_words.index(w)] += 1

        return 1 - cosine_distance(vector1, vector2)
```

We can learn to use this method checking the test:

```
    def test_sentencesimilarity_method_returns_correct_result(self):
        dn = DameNLTK()
        self.assertTrue(dn.sentence_similarity(
        "This is a good sentence".split(),
        "This is a bad sentence".split()) > 0.6)
```

Remember execute the test to learn if the test is running:

```
$ nosetest3 test/test_sentence_similarity.py
```

10.4 Stopwords

For some search engines, these are some of the most common, short function words, such as the, is, at, which, and on. In this case, stop words can cause problems when searching for phrases that include them

You can find the test:

```
$ nosetest3 test/test_stopwords.py
```

10.5 Manage words

Managing words is about definitions, singulars, plurals, synonims and antonims

We can start building singulars and plurals from a word. You can read the test file:

```
import unittest
import nltk
from nltk.stem import *
from nltk.stem.snowball import SnowballStemmer

class TddInPythonExample(unittest.TestCase):
```

```
    def test_stem_returns_correct_result(self):
        stemmer = PorterStemmer()
        self.assertEqual(stemmer.stem("vuelos"), "vuelo")

    def test_snowball_stem_returns_correct_result(self):
        sb = SnowballStemmer("english").stem("generously")
        self.assertEqual(sb, "generous")
```

You can test it with:

```
$ nosetest3 test/test_stem.py
```

Another tool for NLP tasks is to retrieve synonims and antonims from words. You can execute the tests:

```
$ nosetest3 test/test_syn.py
```

Another way is using wordnet:

```
$ nosetests3 test/test_wordnet.py
```

Wordnet provides measures about semantic similarity between words, definitions about words, ...

10.6 Corpus

The corpus allows use data very useful in NLP (ex: gutemberg files, twitter examples, wordnet, names with gender, ...). You can check the source to learn how to use it:

```
$ nosetests3 test/test_corpus.py
$ nosetests3 test/test_gutenberg.py
```

10.7 Gramatical Trees

The gramatical trees are a tool in Natural Language Processing to draw the gramatical in a visual or computing way. We have created tests with gramatical trees:

```
$ nosetests3 test/test_semantic_tree.py
```

11 DameWebutils: Checking images and urls

You can use python testing in Django or Flask on a similar way than in another libraries, but there are some tasks as check broken urls or images that you can use in any web site. That's the subject of this chapter.

Now, you can install damewebutils easily in a python virtual environment:

```
davidam-Lenovo-B50-10:~/git/damewebutils$ mkdir /tmp/venv
davidam-Lenovo-B50-10:~/git/damewebutils$ python3 -m venv /tmp/venv/
davidam-Lenovo-B50-10:~/git/damewebutils$ cd /tmp/venv/
davidam-Lenovo-B50-10:/tmp/venv$ ls
bin  include  lib  lib64  pyvenv.cfg  share
davidam-Lenovo-B50-10:/tmp/venv$ source bin/activate
(venv) davidam-Lenovo-B50-10:/tmp/venv$ pip3 install damewebutils
Processing /home/davidam/.cache/pip/wheels/6a/92/21/f0210ab109dd30d573ec84c79c7a2cc556
0.1.2-py3-none-any.whl
Collecting markdown
  Downloading Markdown-3.3.4-py3-none-any.whl (97 kB)
     || 97 kB 565 kB/s
Collecting requests
  Using cached requests-2.25.1-py2.py3-none-any.whl (61 kB)
Collecting lxml
  Using cached lxml-4.6.2-cp38-cp38-manylinux1_x86_64.whl (5.4 MB)
Collecting cssselect
  Using cached cssselect-1.1.0-py2.py3-none-any.whl (16 kB)
Collecting idna<3,>=2.5
  Using cached idna-2.10-py2.py3-none-any.whl (58 kB)
Collecting certifi>=2017.4.17
  Using cached certifi-2020.12.5-py2.py3-none-any.whl (147 kB)
Collecting urllib3<1.27,>=1.21.1
  Using cached urllib3-1.26.3-py2.py3-none-any.whl (137 kB)
Collecting chardet<5,>=3.0.2
  Using cached chardet-4.0.0-py2.py3-none-any.whl (178 kB)
Installing collected packages: markdown, idna, certifi, urllib3, chardet, re-
quests, lxml, cssselect, damewebutils
Successfully installed certifi-2020.12.5 chardet-4.0.0 cssselect-1.1.0 damewebutils-
0.1.2 idna-2.10 lxml-4.6.2 markdown-3.3.4 requests-2.25.1 urllib3-1.26.3
```

Now you can execute dameimgs.py and dameurls.py giving some url as argument. For example:

```
(venv) davidam-Lenovo-B50-10:/tmp/venv$ dameimgs.py https://www.davidam.com
https://www.davidam.com
https://www.davidam.com/img/davidam-pypi.jpg
200
https://www.davidam.com/img/davidam-pypi.jpg
200
Imgs with possible troubles in []
```

In this example, we have found one image using the html markup on a right way.

With the urls is similar:

```
$ dameurls.py https://www.davidam.com
https://twitter.com/davidam9
https://scholar.google.es/citations?user=znKwqIMAAAAJ&hl=es&oi=ao
https://orcid.org/0000-0002-2986-5361
https://code.orgmode.org/bzg/org-mode/src/master/contrib/lisp/org-effectiveness.el
https://code.orgmode.org/bzg/org-mode/src/master/contrib/lisp/org-license.el
https://savannah.nongnu.org/projects/drupal-el
https://savannah.nongnu.org/projects/php-ext-el
http://drupal.org/project/orgmode
http://drupal.org/project/ocrad
https://github.com/davidam/damegender
[...]
Urls with troubles:
['https://github.com/davidam/damegender/blob/dev/manual/damegender.es.pdf']
```

12 DameDB

12.1 SQLite

SQLite is the most simple SQL database. You can create and check a sql datamodel executing the next tests:

```
$ nosetests3 tests/test_damedb_sqlite.py
```

12.2 MySQL

MySQL is the most used free sql database.

You need create a database model with:

```
$ mysql -u root < files/create.sql
```

Later you can check python operation with the database:

```
$ nosetests3 tests/test_damedb_mysql.py
```

12.3 Mongo

Mongo is a nosql database very used in the market.

The first step to use Mongo in Python is to install Mongo in a computer. You can read the manual to install mongo in different operating systems `https://docs.mongodb.com/manual/tutorial/`.

The second step is to install python libraries about mongo. For example, mongoengine.

```
python -m pip install mongoengine
```

Later you can make exercises about mongo in python. Run your tests!

```
$ nosetests3 tests/test_damedb_mongo.py
```

Appendix A License

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0. PREAMBLE

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